

1 Solution — GIAM 5.3

Problem 5

1.1 Problem

Give an inductive proof that $\forall n \in \mathbb{N}, 6 \mid 3n^2 + 3n - 12$.

1.2 Answer

$$3(k+1)^2 + 3(k+1) - 12 = (3k^2 + 3k - 12) + 6(k+1)$$

The step adds a visible multiple of 6, so divisibility passes from k to $k+1$. Verified for every n up to 5000.

1.3 The Proof

Write $f(n) = 3n^2 + 3n - 12$.

Theorem. For every natural number n , 6 divides $f(n)$.

Proof. By induction on n .

Basis. At $n = 0$, $f(0) = -12 = 6 \cdot (-2)$, and -2 is an integer, so $6 \mid f(0)$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $f(k) = 6m$ for some integer m . Expanding,

$$f(k+1) = 3k^2 + 6k + 3 + 3k + 3 - 12 = (3k^2 + 3k - 12) + 6k + 6.$$

So $f(k+1) = f(k) + 6(k+1)$, and substituting the hypothesis,

$$f(k+1) = 6m + 6(k+1) = 6(m+k+1).$$

Since $m+k+1$ is an integer, $6 \mid f(k+1)$.

Basis and step together give the result for every n . ■

The basis value is negative, and that is fine: $d \mid a$ means $a = dq$ for some *integer* q , and negative integers count. The first two values, $f(0) = -12$ and $f(1) = -6$, are both multiples of 6.

1.4 The Picture

The values grow by 6, 12, 18, ..., and each one is six times a number two less than a triangular number, as drawn below.

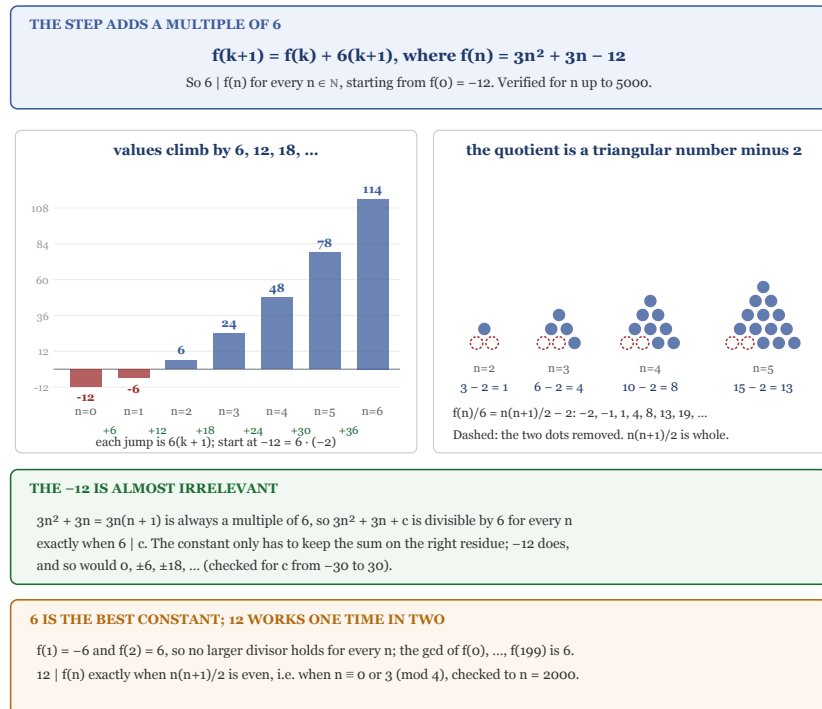


Figure 1.1: A diagram in four parts. The top band states the step identity f of k plus one equals f of k plus six times k plus one, where f of n is three n squared plus three n minus twelve. The left panel is a bar chart of f of n for n from zero to six: minus twelve, minus six, six, twenty-four, forty-eight, seventy-eight, one hundred fourteen, with the jumps six, twelve, eighteen, twenty-four, thirty, thirty-six marked between bars. The right panel shows triangles of dots for n from two to five, with three, six, ten and fifteen dots, two of each drawn dashed as removed, leaving one, four, eight and thirteen: the quotient f of n over six. A green band notes that three n squared plus three n plus c is divisible by six for every n exactly when six divides c . An

orange band notes that six is the best constant, and that twelve divides f of n exactly when n is congruent to zero or three mod four.

1.5 Remark — Why This Step Is Easier Than Problems 1 and 2

In 5.3 Problem 1 the step added $3k(k+1)$, and in Problem 2 it added $3(k^2+k+2)$. Both are obviously multiples of 3, but the sharper remarks there, that 6 divides as well, needed the extra fact that $k(k+1)$ is even. Here the increment $6(k+1)$ is a multiple of 6 on its face, so the induction needs nothing else.

The reason is a general feature of quadratics. For any a, b ,

$$(a(k+1)^2 + b(k+1)) - (ak^2 + bk) = 2ak + (a+b),$$

checked symbolically and for k from -100 to 99 . With $a = b = 3$ the difference is $6k + 6$. Both terms carry a factor 6 because $2a = 6$ and $a + b = 6$. This first difference is the discrete analogue of a derivative, the same finite differences used in 5.2: a quadratic always has linear first differences.

1.6 Remark — The Direct Proof: Triangular Numbers

Factoring gives the result without induction:

$$f(n) = 3n(n+1) - 12 = 6 \left(\frac{n(n+1)}{2} - 2 \right) = 6(T_n - 2),$$

where $T_n = 1 + 2 + \cdots + n$ is the n -th triangular number. T_n is a whole number, being a sum of integers; equivalently, one of n and $n+1$ is even. The identity was checked for n up to 5000. The quotients are $-2, -1, 1, 4, 8, 13, 19, \dots$, which the right panel of the figure shows as triangles with two dots removed.

The factored form also proves the statement for every *integer* n , not only natural numbers, because $n(n+1)$ is even for negative n as well. In fact $f(-n-1) = f(n)$: the parabola is symmetric about $n = -\frac{1}{2}$, so the negative integers repeat the natural-number values. Both facts were checked for $|n|$ up to 3000.

1.7 Remark — Is 6 the Best Divisor?

Yes. Since $f(1) = -6$, no constant larger than 6 can divide every value, and the gcd of $f(0), \dots, f(199)$ is 6. As in 5.3 Problem 3, the stated divisor is already sharp.

The constant -12 plays a small role. Because $3n^2 + 3n$ is always a multiple of 6,

$$6 \mid 3n^2 + 3n + c \text{ for every } n \iff 6 \mid c,$$

and a search over c from -30 to 30 finds exactly the multiples of 6. So -12 could be swapped for $0, \pm 6, 18$, and so on, with the same proof.

Along a subsequence there is more. $12 \mid f(n)$ exactly when $T_n - 2$ is even, that is, when T_n is even. The triangular numbers $0, 1, 3, 6, 10, 15, 21, 28, \dots$ are even precisely for $n \equiv 0$ or $3 \pmod{4}$, so 12 divides $f(n)$ for half of all n , in the repeating pattern yes, no, no, yes. This was checked to $n = 2000$.